

## Getting started with STiRoT (ST immutable Root of Trust) for STM32H5 MCUs

### Introduction

When deploying a device in an untrusted environment, it is important to consider the potential threats that may compromise the device's security.

To mitigate these risks, it is recommended to only allow authentic firmware to run on the device.

Firmware updates are common for connected devices to fix bugs, introduce new features, or deploy new security improvement. However, it is important to install these updates securely to ensure the integrity of the device. Without proper security measures, firmware updates can result in serious consequences such as firmware cloning, malicious software download, or device corruption. To protect sensitive data and critical operations, security solutions must be designed leveraging cryptography and memory protections.

Cryptography ensures the authenticity and confidentiality of firmware update, while memory protection mechanisms prevent unauthorized access to the device.

This application note gives an overview of the STiRoT solution integrated in Arm® TrustZone® STM32H5 microcontrollers, based on the Arm® Cortex®-M33 processor, with its associated tools ecosystem. How to use it, step by step, is described at [5].

**Table 1. Applicable products**

| Type            | Products                                                                                                                              |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Microcontroller | STM32H5F5LJ, STM32H5F5ZJ, STM32H5F5VJ, STM32H5F5J-DK, STM32H5F5JDIE, STM32H5F5IJ                                                      |
|                 | STM32H553ZG, STM32H553VG, STM32H553UG, STM32H553RG, STM32H553CG                                                                       |
|                 | STM32H573RI, STM32H573OI, STM32H573VI, STM32H573ZI, STM32H573II, STM32H573AI, STM32H573MI, STM32H573I-DK, STM32H573IIU, STM32H573IDIE |
|                 | STM32H533ZE, STM32H533VE, STM32H533RE, STM32H533HE, STM32H533CE                                                                       |

## 1 General information

This document applies to the **STM32H5 series** Arm® Cortex®-M33-based microcontrollers.

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**Table 2. Terms and abbreviations**

| Acronym | Definition                                    |
|---------|-----------------------------------------------|
| BL      | Bootloader                                    |
| HDPL    | Hardware protection level                     |
| MCU     | Microcontroller unit                          |
| MPU     | Memory protection unit                        |
| OB      | Option bytes                                  |
| OBKeys  | Option bytes keys                             |
| SBSFU   | Secure boot and secure firmware update        |
| SESIP   | Security evaluation standard for IOT platform |
| STiRoT  | ST immutable root of trust                    |
| uRoT    | Updatable root of trust                       |
| SDP     | Secure data provisioning                      |
| TLV     | Type length value                             |
| TPC     | STM32 Trusted Package Creator                 |

### Reference documents

**Table 3. List of documents**

| Reference | Name/address                                                                                                                                                                    | Title                                                                                            |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| [1]       | AN6008                                                                                                                                                                          | <i>Introduction to Debug Authentication (DA) for STM32 MCUs</i> application note                 |
| [2]       | RM0517                                                                                                                                                                          | <i>STM32H5E4/H5F4 and STM32H5E5/H5F5 Arm®-based 32-bit MCUs</i> reference manual                 |
| [3]       | <a href="#">RM0481</a>                                                                                                                                                          | <i>STM32H56x/STM32H573xx and STM32H523xx/STM32H533xx Arm®-based 32-bit MCUs</i> reference manual |
| [4]       | RM0539                                                                                                                                                                          | <i>STM32H543/553xx Arm®-based 32-bit MCUs</i> reference manual                                   |
| [5]       | <a href="https://wiki.st.com/stm32mcu/wiki/Category:How_to_start_with_STiRoT_on_STM32H5">https://wiki.st.com/stm32mcu/wiki/Category:How_to_start_with_STiRoT_on_STM32H5</a>     | <i>How to start with STiRoT on STM32H5</i> wiki article <sup>(1)</sup>                           |
| [6]       | <a href="https://wiki.st.com/stm32mcu/wiki/Security:How_to_start_with_STiRoT_on_STM32H573">https://wiki.st.com/stm32mcu/wiki/Security:How_to_start_with_STiRoT_on_STM32H573</a> | <i>How to start with STiRoT on STM32H573</i> wiki article <sup>(1)</sup>                         |
| [7]       | AN6014                                                                                                                                                                          | Introduction to SFSP versions for STM32H5 MCUs                                                   |

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**Table 4. External references**

| Reference | Name/address                                        | Title                  |
|-----------|-----------------------------------------------------|------------------------|
| [8]       | <a href="http://mcuboot.com">http://mcuboot.com</a> | MCUBoot <sup>(1)</sup> |

1. This URL belongs to a third-party. It is active at document publication. However, STMicroelectronics shall not be liable for any change, move, or inactivation of the URL or the referenced material.

## 2 STiRoT presentation

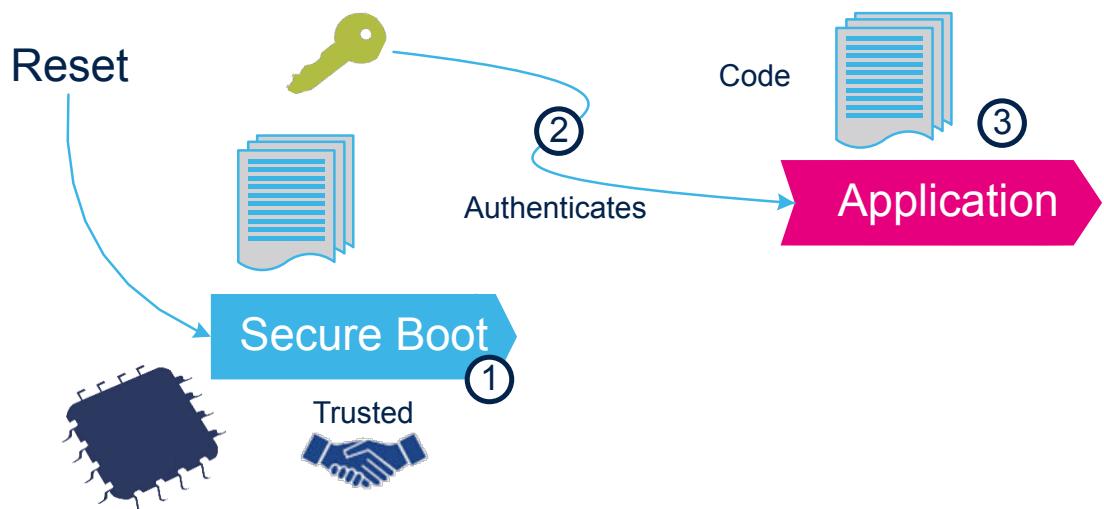
### 2.1 Overview

STiRoT stands for **ST** immutable (unchangeable) **Root of Trust**, and acts as the first boot stage. STiRoT targets a SESIP level 3 certified implementation.

STiRoT is embedded in an immutable area of the STM32H5 and provides two services:

- The Secure Boot (root of trust services) is an immutable code, which is always executed after a system reset (1). It activates runtime protections and then, it verifies the authenticity and integrity of the application (2) code before every execution (3).

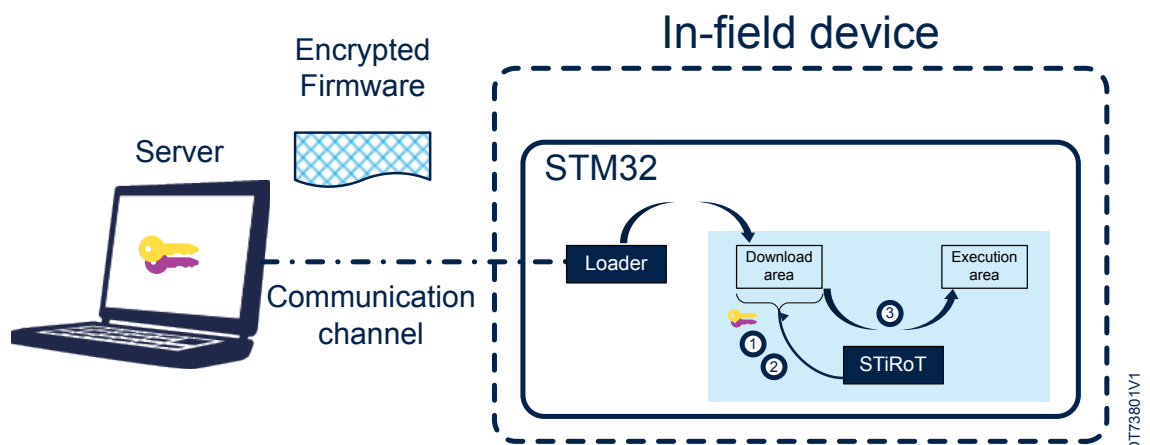
Figure 1. Secure Boot



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- The Secure Firmware Update application is an immutable code that detects that a new firmware image is available. It verifies the image authenticity (1), integrity (2) and finally installs the decrypted firmware (3).

Figure 2. Secure Firmware Update



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The detailed startup sequence is described in [Appendix A: STiRoT start-up sequence](#) as well as the most efficient way of working during development phase in [Appendix B: Application development phase](#).

## 2.2 Protection measures and security strategy

Cryptography ensures integrity, authentication, and confidentiality. However, the use of cryptography alone is not enough: a set of hardware and software security measures are supported by the STiRoT for protecting critical operations and sensitive data (such as a secret key). The SESIP level 3 certification is targeted.

The following cryptographic scheme is implemented in STiRoT:

- ECDSA-P256 asymmetric cryptography for image authentication.
- AES-CTR-128 symmetric cryptography with key ECIES-P256 encrypted for image confidentiality.
- SHA256 cryptography for image integrity check.
- All cryptography operations are hardware-accelerated.

## 2.3 STiRoT activation

On STM32H5, STiRoT activation is done by:

1. Configuring the `BOOT_UBE` option byte (refer to [3] or [2] for more details) in order to boot on STiRoT.
2. Defining STiRoT configuration. The main parameters are:
  - Location, size of each area hosting user application code and data images in user flash memory.
  - Authentication and encryption keys used by STiRoT.

Section 4 provides details on all configuration parameters.

Both operations are part of the provisioning process.

STiRoT is activated in two different use cases:

- **One boot stage:** The STiRoT integrated inside the STM32H5 manages directly the user application.
- **Two boot stages:** The STiRoT integrated inside the STM32H5 manages an updatable boot stage (uRoT) located in the user flash memory which manages the user application. The updatable boot stage can be customized to fit customer needs.

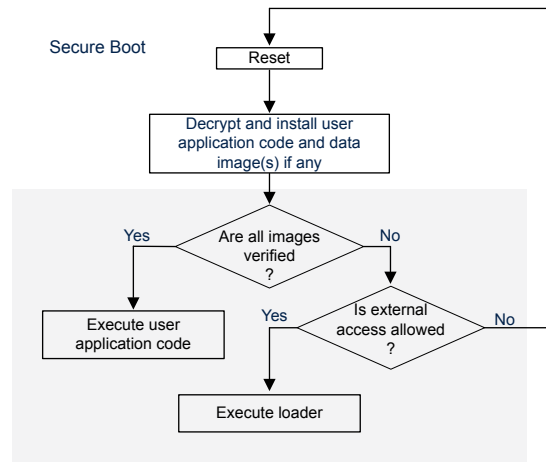
**Table 5. Boot time performance**

| Configuration      | Boot time  |            |          |
|--------------------|------------|------------|----------|
|                    | 256 Kbytes | 512 Kbytes | 1 Mbytes |
| -                  |            |            |          |
| Execution @ 64MHz  | 41,5 ms    | 70 ms      | 128 ms   |
| Execution @ 200MHz | 18 ms      | 27,5ms     | 46ms     |
| Execution @ 250MHz | 16ms       | 23,5ms     | 38ms     |

### 2.3.1 One boot stage

At reset, STiRoT is executed in temporal isolation 1 (HDPL1). Refer to [2] or [3] for more details on temporal isolation levels. After a successful verification of the authenticity and the integrity of the user application code and data images, STiRoT execute the user application in secure mode (HDPL2). In case of verification failure, STiRoT jumps into the bootloader to provide the end user a way to download a new user application code and/or data images if the external access is configured as allowed.

Figure 3. STiRoT behavior in case of verification failure



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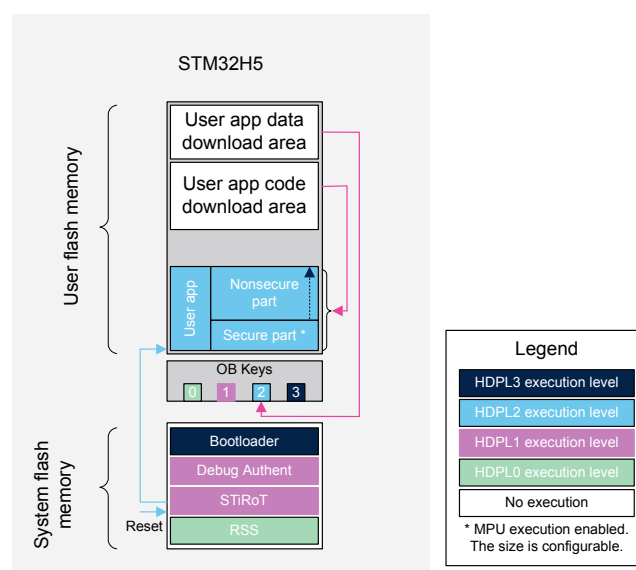
The user application can be fully secure or divided in a secure and nonsecure application. In both cases, the image stored in the download area is a single image. The secure part of the user application must contain at least the vector table and the reset handler. If this is not the case, the application does not start as STiRoT executed it in secure mode.

The secrets (data) associated to the user application are installed by STiRoT in the protected HDPL2 OBKeys. When no secrets are required by the application, the data image management can be deactivated through STiRoT configuration.

To ensure the security of the device, the MPU is configured to allow only the application code area to be executed when STiRoT jumps into application, minimizing the risk of unauthorized code execution. It is the user application responsibility to reconfigure the secure MPU to fit with its security needs.

The following figure gives an example of the memory mapping when STiRoT is activated in a one boot stage configuration.

Figure 4. STM32H5 - STiRoT - User application executed after STiRoT



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An example of one boot stage application is provided with STM32CubeH5 MCU Package.

**Note:** When required, this is the user application responsibility to move to HDPL3 and to switch in nonsecure mode.

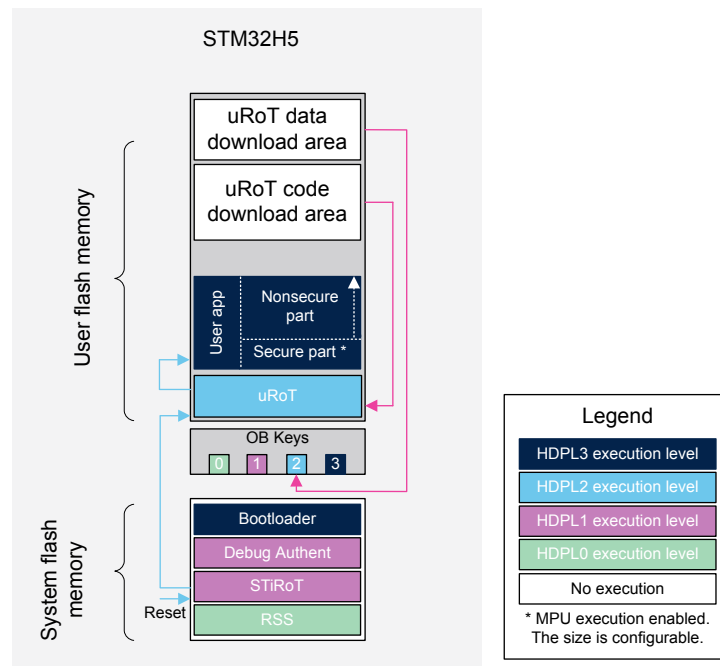
### 2.3.2 Two boot stages

The uRoT (updatable Root of Trust) located in the user flash memory act as a second boot stage. At reset, STiRoT is executed in temporal isolation 1, HDPL1. Refer to [2] or [3] or more details on temporal isolation levels. After a successful verification of the authenticity and the integrity of uRoT code and data images, STiRoT executes uRoT in HDPL2 and in secure mode. Then, uRoT verify the user application before executing it. In case of verification failure of uRoT code and/or data image, STiRoT jumps into the bootloader to provide the end user a way to download a new uRoT code and/or data images if the external access is configured as allowed. See [Figure 3. STiRoT behavior in case of verification failure](#) in case of verification failure.

The uRoT is a fully secure application. The secret keys managed by uRoT and hosted in the data image are installed by STiRoT in the protected HDPL2 OBKeys.

To ensure the security of the device, the MPU is configured to allow only the uRoT code area to be executed when STiRoT jumps into uRoT, minimizing the risk of unauthorized code execution. It is the uRoT responsibility to reconfigure the secure MPU to fit with its security needs. In this case, the security of the user application depends on the uRoT strategy.

**Figure 5. STM32H5 - STiRoT - User application executed after uRoT**



An example of a two boot stages application is provided with the STM32H5 series MCU Package.

## 2.4 STiRoT constraints

The list of constraints to be taken into account when activating the STiRoT boot path is the following:

- The independent watchdog (IWDG) peripheral offers a high safety level, thanks to its capability to prevent malfunctions due to software or hardware failures in a disturbed environment. For application requesting such high safety level, it is recommended to activate IWDG via option bytes when using STiRoT and regularly reload the IWDG on the application side.
- WWDG activation by option byte is not supported by STiRoT. Only IWDG can be activated through option byte.
- STiRoT does not support potential mode for the tamperers. It resets in loop until power-off if tamper event in potential mode occurs during its execution (or during Standby mode or VBAT mode).
- When a tamper event occurs during STiRoT execution (or Standby mode or VBAT mode), then STiRoT saves the content of the TAMP status register, and write a magic value at the end of backup SRAM. The application can check the presence of this magic value and stored status register in backup SRAM to identify which tamper event(s) occurred, then decides the appropriated action and clears the magic value.

*Note: This is applicable from the SFSP version 2.7.0 on STM32H573xx products , the SFSP version 1.3.0 on STM32H533xx products , the SFSP version 1.3.0 on STM32H553xx products, and SFSP version 1.1.0 on STM32H5F5xx products. For previous SFSP version, when a tamper fault occurs, the system resets in loop until a backup domain power-on reset is performed. Refer to document [7] to get all the information about SFSP version tracking.*

- Internal tamper 9 (fault generation for cryptographic peripherals (SAES, PKA, AES, RNG) and internal tamper 15 (system fault detection) are activated by default during STiRoT execution and remains activated when jumping into the user application. The usage of these tamperers in STiRoT can be deactivated through STiRoT OBKeys configuration (see [Section 4: OBKeys configuration file](#)).

*Note: This is applicable from the SFSP version 2.7.0 on STM32H573xx products , the SFSP version 1.3.0 on STM32H533xx products , the SFSP version 1.3.0 on STM32H553xx products, and SFSP version 1.1.0 on STM32H5F5xx products. For previous SFSP version, when a tamper fault occurs, the system resets in loop until a backup domain power-on reset is performed. Refer to document [7] to get all the information about SFSP version tracking.*

- MPU is configured to only allow the user application code area to be executed when STiRoT jumps into the user application. It is the user application responsibility to reconfigure the MPU to fit with its security needs.
- SRAM2 is used by STiRoT and is therefore erased at each reset.
- SWAP\_BANK activation by option byte is not supported by STiRoT.

**Table 6. STM32H5F5xx and STM32H573xx backup SRAM mapping**

| Backup SRAM mapping | Size (bytes) | Description                                                           |
|---------------------|--------------|-----------------------------------------------------------------------|
| 0x5003 73FC         | 0x4          | Tamper status register copy (indicates which tamper(s) have occurred) |
| 0x5003 73F8         | 0x4          | Magic value (if a tamper fault occurred, it equals 0x003981BB)        |

**Table 7. STM32H553xx and STM32H533xx backup SRAM mapping**

| Backup SRAM mapping | Size (bytes) | Description                                                           |
|---------------------|--------------|-----------------------------------------------------------------------|
| 0x5003 6BFC         | 0x4          | Tamper status register copy (indicates which tamper(s) have occurred) |
| 0x5003 6BF8         | 0x4          | Magic value (if a tamper fault occurred, it equals 0x003981BB)        |



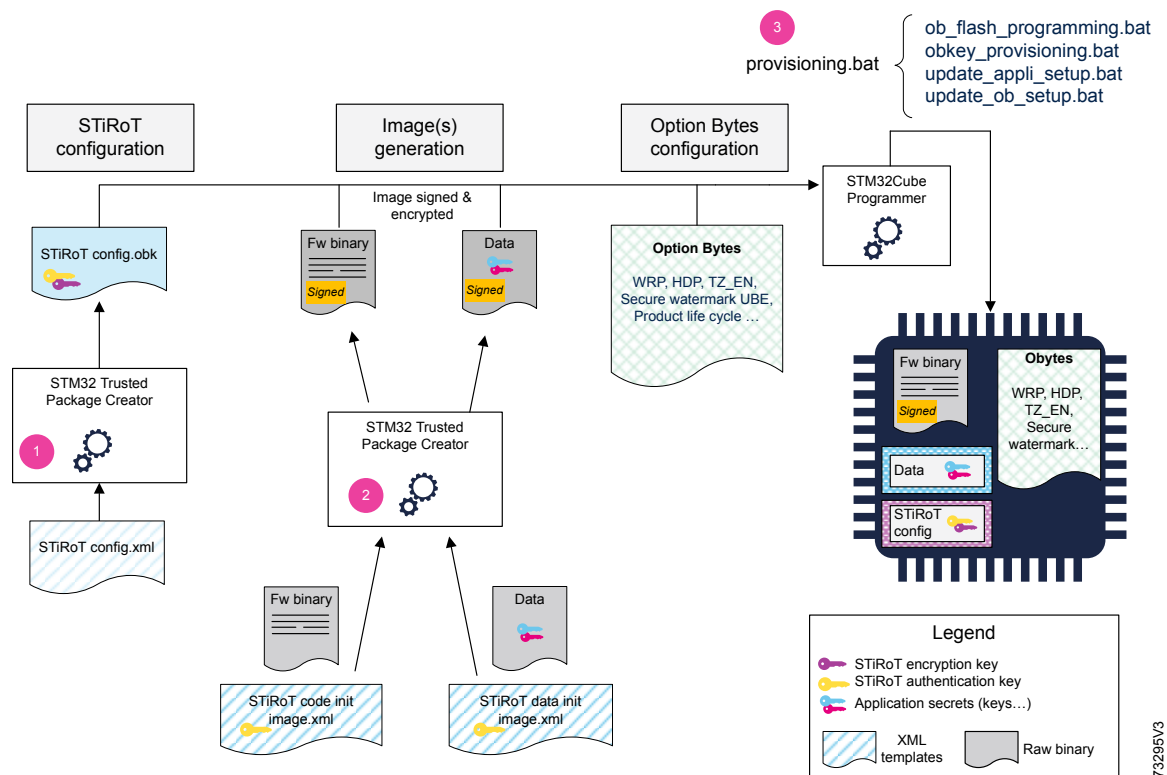
### 3 Provisioning process

The product provisioning to activate and configure STiRoT is done following the three steps below when executing provisioning.bat:

1. Configuration of the STiRoT. At this stage, the number of images managed (firmware image only, or firmware and data images), the location of the images, and the cryptographic keys are defined.
2. Generation of the image(s).
3. Programming of the option bytes, OBKeys, and image(s) in the device.

**Note:** A set of scripts is provided in the STM32CubeH5 FW package (Firmware/Projects/Board/ROT\_Provisioning/STiRoT folder). It guides the user all along the provisioning process. Refer to [6] for more information.

Figure 6. STiRoT provisioning process



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## 4 OBKeys configuration file

STiRoT OBKeys configuration file (STiRoT\_Config.obk) is generated using STM32 Trusted Package Creator with a template file listing all the different parameters (STiRoT\_Config.xml) as input.

STiRoT configuration provides the possibility to:

- Define the user flash memory mapping: select the location and the size of each area.
- Configure the authentication and encryption keys.
- Activate the optional management of a data image.
- Allow the jump in the bootloader (external access) when no valid image is installed.

Some parameters are considered as advanced configuration, and are therefore hidden when the STiRoT configuration is displayed by STM32 Trusted Package Creator. If required, the `<Hidden>` `</Hidden>` property can be switched from 1 to 0 in the STiRoT\_Config.xml file.

The list of all the parameters is the following:

**Table 8. STiRoT Configuration**

| Parameter                          | Hidden | Description                                                                                                                                                                                                                                                                                                                                                 | Additional controls/constraints                                                                                                                                                                                                                                                                                                               |
|------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of images managed by STiRoT | No     | <ul style="list-style-type: none"> <li>• 1: Firmware image only. Configuration selected for any application which does not manage secret information.</li> <li>• 2: Firmware and data images. As example, in the two boot stages use case, the data image contains the cryptographic keys managed by uRoT application.</li> </ul>                           | Slot definition of data image (size and offset) will be available through STM32 Trusted Package Creator as soon as two images configuration is selected.                                                                                                                                                                                      |
| High speed clock activation        | Yes    | <ul style="list-style-type: none"> <li>• 0: 64 MHz, initial clock configuration</li> <li>• 1: 200 MHz (default value), best speed clock supporting the full range of temperature</li> <li>• 2: 250 MHz, max speed clock</li> </ul>                                                                                                                          | -                                                                                                                                                                                                                                                                                                                                             |
| Is the firmware full secure        | No     | <ul style="list-style-type: none"> <li>• Yes: The firmware booting is fully secure. Refer to the example <code>STiRoT_Appli</code> provided in STM32H5 series Cube.</li> <li>• No: The firmware booting is made of a secure part and a nonsecure part. Refer to the example <code>STiRoT_Appli_TrustZone</code> provided in STM32H5 series Cube.</li> </ul> | The size of the secure area inside the firmware execution area must be specified as soon as the firmware is not fully secure. In this configuration, the image generation triggered by the post build command of the nonsecure project is preceded by an assembly command of the secure and nonsecure binaries.                               |
| Jump into ST bootloader            | Yes    | <ul style="list-style-type: none"> <li>• Yes (default value): STiRoT jump into the ST bootloader when no valid firmware image is installed.</li> <li>• No: Jump into the ST bootloader is not allowed.</li> </ul>                                                                                                                                           | When a jump is not allowed, a reset in loop is generated when no valid firmware image is installed.                                                                                                                                                                                                                                           |
| Firmware execution area offset     | No     | The offset is from the beginning of the user flash memory.<br>It must be aligned on a sector start address.                                                                                                                                                                                                                                                 | No overlap is allowed between the execution areas and the download areas.<br>A firmware execution area cannot be mapped between the two download areas.                                                                                                                                                                                       |
| Firmware download area offset      | No     | The offset is from the beginning of the user flash memory.<br>It must be aligned on a sector start address.                                                                                                                                                                                                                                                 | No overlap is allowed between the execution areas and the download areas.<br>A firmware execution area cannot be mapped between the two download areas.<br>Both firmware and data download areas can overlap. In this case, the installation of a data and firmware image in a single execution using the dependency feature is not possible. |
| Firmware area size                 | No     | Must be a multiple of the sector size.                                                                                                                                                                                                                                                                                                                      | No overlap is allowed between the execution areas and the download areas.<br>A firmware execution area cannot be mapped between the two download areas.                                                                                                                                                                                       |

| Parameter                                                  | Hidden | Description                                                                                                                                                                                                                                         | Additional controls/constraints                                                                                                                                                                                                                             |
|------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data area offset in HDPL2 OBKeys                           | No     | The offset is from the beginning of the HDPL2 OBKeys area (0xFFD0900).                                                                                                                                                                              | <ul style="list-style-type: none"> <li>The range allowed is from 0xFFD0900 to 0xFFD0BF0: there is no overlap with the HDPL3 OBKeys.</li> <li>This parameter is disabled for the "one image configuration".</li> </ul>                                       |
| Data slot size in HDPL2 OBKeys                             | No     | The maximum is 0x2F0.                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>The range allowed is from 0xFFD0900 to 0xFFD0BF0: there is no overlap with HDPL3 OBKeys.</li> <li>This parameter is disabled for the "one image configuration".</li> </ul>                                           |
| Data download area offset                                  | No     | The offset is from the beginning of the user flash memory.<br>It must be aligned on a sector start address.                                                                                                                                         | Both firmware and data download areas can overlap. In this case, the installation of a data and firmware image in a single execution using the dependency feature is not possible.                                                                          |
| Data download slot size                                    | Yes    | Must be a multiple of the sector size.                                                                                                                                                                                                              | Must be 0x2000 for STM32H5 series product (parameter hidden).                                                                                                                                                                                               |
| Size of the secure area inside the firmware execution area | No     | Must be a multiple of the sector size.                                                                                                                                                                                                              | This parameter is disabled for the "firmware fully secure" configuration.<br>STiRoT configures the MPU to allow execution only for the secure part of the firmware. Then, it is up to the secure application to reconfigure the MPU depending on its needs. |
| SRAM2 erasing in case of reset                             | Yes    | Must be yes: the SRAM2 is automatically erased by hardware in case of reset. It ensure that all secrets are erased in case of reset.                                                                                                                | This information is duplicated to ensure that the option bytes are correctly configured.                                                                                                                                                                    |
| SRAM2 ECC management activation                            | Yes    | <ul style="list-style-type: none"> <li>Yes (default value): a reset is generated in case of ECC detection. ECC is a mechanism to detect memory tampering.</li> <li>No: ECC is not managed when SRAM tampering detection is not required.</li> </ul> | This information is duplicated to ensure that the option bytes are correctly configured.                                                                                                                                                                    |
| Product state minimal allowed                              | No     | Open, Provisioning, Provisioned, TZ-Closed, Closed, Locked: STiRoT only execute the firmware if the product state programmed in the option byte is equal or greater than the "product state minimal allowed".                                       | If the user plans to use partial regression, then they must set the product state minimal to TZ-Closed (at maximum).                                                                                                                                        |
| Encryption key                                             | No     | The key used to encrypt the firmware and data images.                                                                                                                                                                                               | When this key is regenerated, both firmware and data images must be regenerated with the STM32 Trusted Package Creator "Image Gen" tab (STiRoT_Code_Image.xml and STiRoT_Data_Image.xml)                                                                    |
| Authentication key                                         | No     | The key used to authenticate the firmware and data images.                                                                                                                                                                                          | When this key is regenerated, both firmware and data images must be regenerated with the STM32 Trusted Package Creator "Image Gen" tab (STiRoT_Code_Image.xml and STiRoT_Data_Image.xml)                                                                    |
| Internal tamper 9 and 15 activation. <sup>(1)</sup>        | Yes    | Yes (default value): Internal tamper 9 and 15 are activated during STiRoT execution.<br>No: Internal tamper 9 and 15 are deactivated during STiRoT execution.                                                                                       | -                                                                                                                                                                                                                                                           |

1. This is applicable from the SFSP version 2.7.0 on STM32H573xx products, the SFSP version 1.3.0 on STM32H533xx products, the SFSP version 1.3.0 on STM32H553xx products, and SFSP version 1.1.0 on STM32H5F5xx products. For previous SFSP version, when a tamper fault occurs, the system resets in loop until a backup domain power-on reset is performed. Refer to document [7] to get all the information about SFSP version tracking.

A detailed presentation of the format of STiRoT\_Config.obk file is provided in [Appendix C: STiRoT Config.obk file format](#).

## 5 Images generation

STiRoT manages images at MCUboot format including:

- A header,
- The encrypted firmware or data binary,
- Metadata informations (TLV format: Tags Length Value) allowing the control of the image (SHA256, encryption key...),
- A magic number to trigger the installation.

Further information about the MCUboot open source software is available at [8].

STM32 Trusted Package Creator is provided to generate both firmware and data images. These images are encrypted and signed using the keys configured in the OBKeys configuration file.

Two types of images are generated:

- Initial image: Image in clear to be programmed in execution area during initial device programming,
- Update image: Encrypted image to be programmed in download area. It is installed during secure firmware update process.

### 5.1 Initial firmware image generation

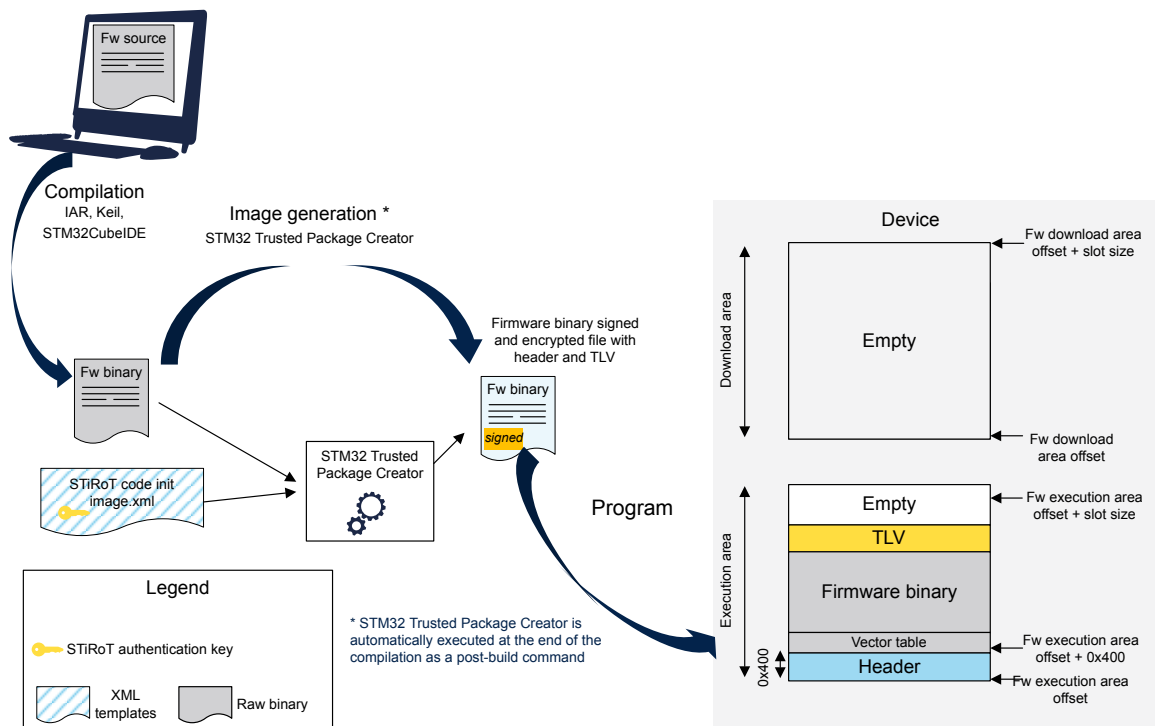
STiRoT\_Code\_Init\_Image.xml contains all the parameters driving the image generation such as:

- The authentication keys.
- The version.

STiRoT\_Code\_Init\_Image.xml can be edited with STM32 Trusted Package Creator in order to modify the version information. The keys are directly inherited from the STiRoT\_Config.xml file.

The following figure shows the firmware image generation:

**Figure 7. Firmware init image generation**



The list of all parameters is detailed in the appendix [Appendix E: Code init image XML file](#).

## 5.2 Update firmware image generation

STiRoT\_Code\_Image.xml contains all the parameters driving the image generation such as:

- The authentication and encryption keys.
- The version and the dependency information.

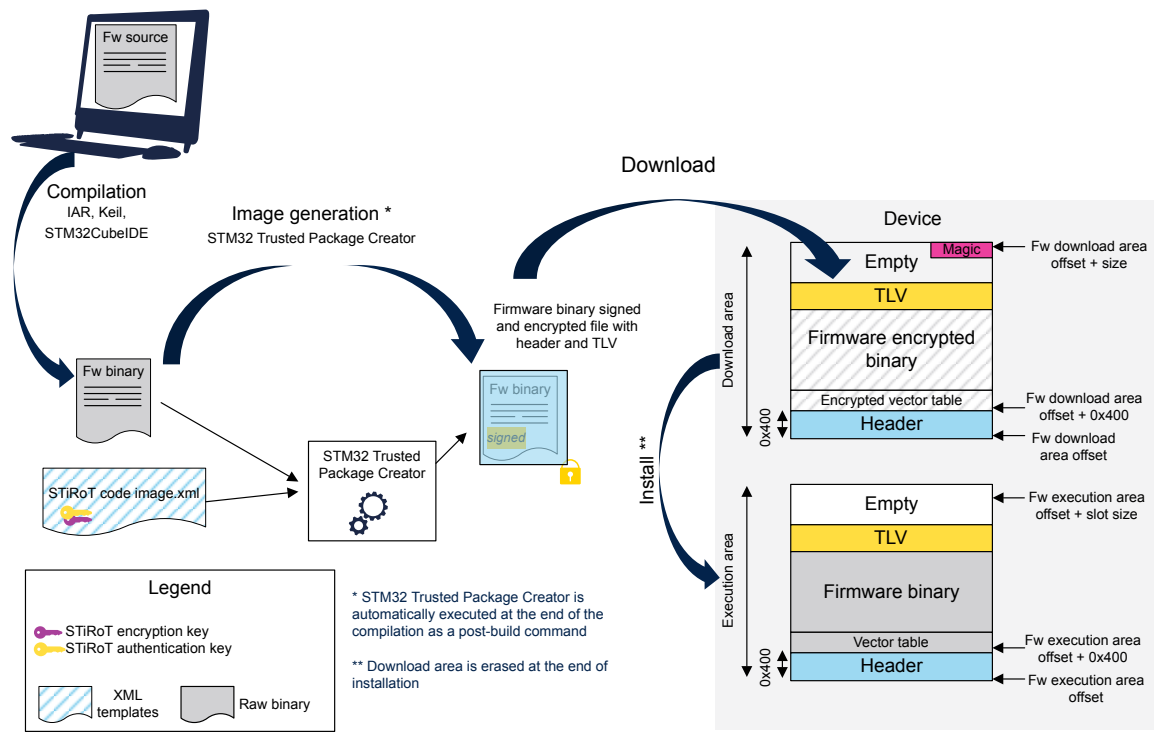
STiRoT\_Code\_Image.xml can be edited with STM32 Trusted Package Creator in order to modify the version and the dependency information. The keys are directly inherited from the STiRoT\_Config.xml file.

Dependency can be enabled when the compatibility with a specific data image version is required: the installation of both images is synchronized in this case.

To generate a clear (non-encrypted) image, the encryption key must be removed from the list of parameters of the xml file.

The following figure shows the firmware image generation:

**Figure 8. Firmware image generation**



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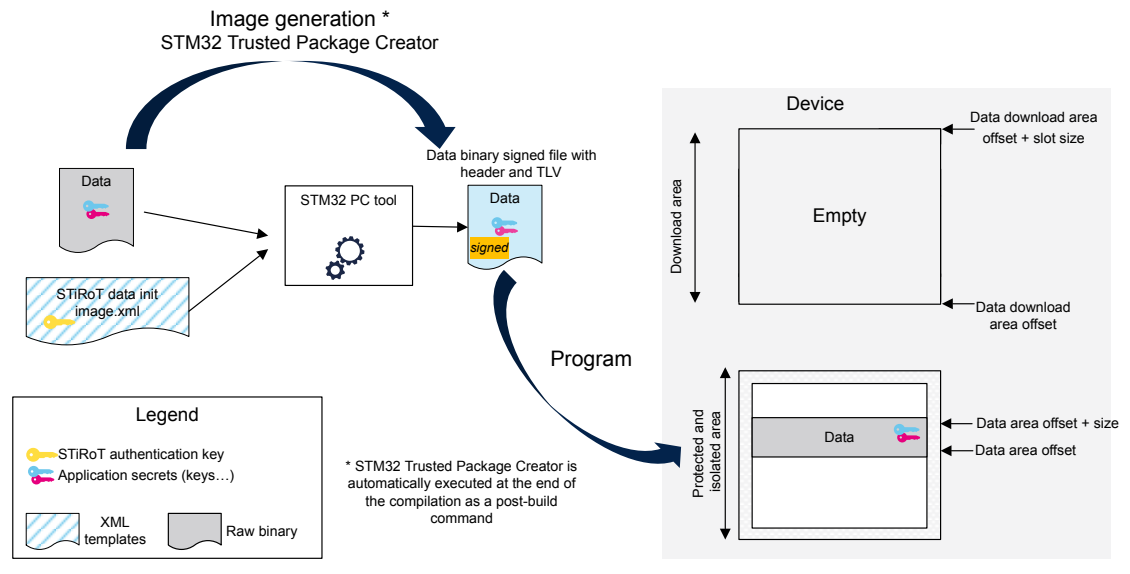
The list of all parameters is detailed in the appendix [Appendix F: Code image XML file](#).

## 5.3 Initial data image generation

The STiRoT\_Data\_Init\_Image.xml file contains all the parameters driving the image generation such as the authentication and the version information. STiRoT\_Data\_Init\_Image.xml can be edited with STM32 Trusted Package Creator in order to modify the version information. The keys are directly inherited from the STiRoT\_Config.xml file.

The following figure shows the data image generation:

Figure 9. Initial data image generation



The list of all parameters is detailed in [Appendix G: Data init image XML file](#).

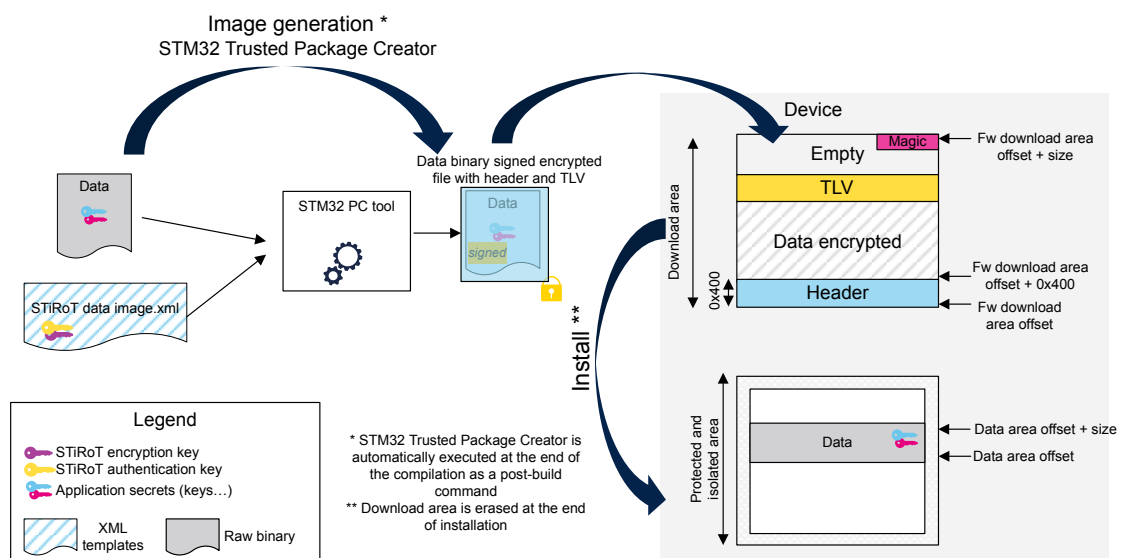
## 5.4 Update data image generation

The STiRoT\_Data\_Image.xml file contains all the parameters driving the image generation such as the authentication and encryption keys, and the version information. STiRoT\_Data\_Image.xml can be edited with STM32 Trusted Package Creator in order to modify the version and the dependency information. The keys are directly inherited from the STiRoT\_Config.xml file.

Dependency can be enabled when the compatibility with a specific firmware image version is required: installation of both images is synchronized in this case

The following figure shows the data image generation:

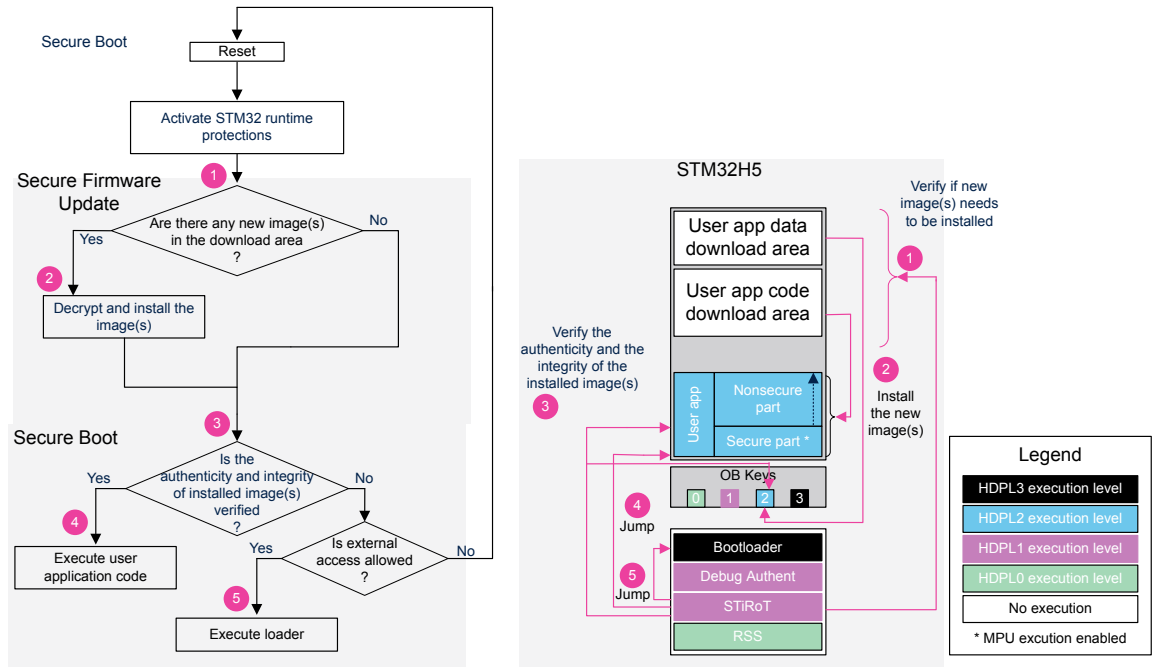
Figure 10. Data image generation



The list of all parameters is detailed in [Appendix H: Data image XML file](#).

## Appendix A STiRoT start-up sequence

Figure 11. STiRoT start-up sequence



The step 5 is optional and can be disabled through STiRoT configuration. When not allowed, a reset is generated instead of a jump to the loader.

## Appendix B Application development phase

The most efficient way to develop and debug an application is to boot directly on the user flash memory in the Open product state. To do this, set the UBE to 0xB4 and the SECBOOTADD to 0x0C000400 using STM32CubeProgrammer (STM32CubeProg). By default, the firmware execution offset is 0x0.

Figure 12. Boot directly in user flash memory

|            |          |            |                                                                                                                                                          |
|------------|----------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| BOOT_UBE   | B4       |            | Unique boot entry control, selects either ST or OEM iRoT for secure boot.<br>B4 : OEM-iRoT (user flash) selected<br>C3 : ST-iRoT (system flash) selected |
| SECBOOTADD | Value    | Address    | Unique Boot Entry Secure Adress                                                                                                                          |
|            | 0xc0004C | 0x0c000400 |                                                                                                                                                          |

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As long as the device is in Open state, the user flash memory can be erased without performing a regression, and a debugger can be connected without performing a Debug Authentication (refer to [1] for more details). Once validated, the Secure Boot path can be activated by configuring STiRoT through the provisioning process.

Refer to [Appendix I: STiRoT execution status](#) execution status for more information on debug information when STiRoT is activated.



## Appendix C STiRoT Config.obk file format

**Note:** Be careful to endianness: `uint32_t 0x01234567` is represented in the obk file as `0x67452301`.

**Table 9. STiRoT\_Config.obk format**

| Offset | Size | Type     | Description                                                                                                                                                             |
|--------|------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | 4    | uint32_t | sdp command header: destination address                                                                                                                                 |
| 4      | 4    | uint32_t | sdp command header: data size address                                                                                                                                   |
| 8      | 4    | uint32_t | sdp command header: encryption enabled/disabled (1/0)                                                                                                                   |
| 12     | 32   | array    | SHA256 of the following data content (from offset 44 to 268)                                                                                                            |
| 44     | 1    | uint8_t  | Number of images managed by STiRoT                                                                                                                                      |
| 45     | 1    | uint8_t  | Clock selection (64/200/250 MHz) – (0/1/2)                                                                                                                              |
| 46     | 1    | uint8_t  | Firmware fully secure (1/0)                                                                                                                                             |
| 47     | 1    | uint8_t  | Jump into ST bootloader enabled/disabled (1/0)                                                                                                                          |
| 48     | 4    | uint32_t | Firmware execution area offset                                                                                                                                          |
| 52     | 4    | uint32_t | Firmware download area offset                                                                                                                                           |
| 56     | 4    | uint32_t | Firmware area size                                                                                                                                                      |
| 60     | 4    | uint32_t | Data area offset in HDPL2 OBKeys                                                                                                                                        |
| 64     | 4    | uint32_t | Data slot size in HDPL2 OBKeys                                                                                                                                          |
| 68     | 4    | uint32_t | Data download area offset                                                                                                                                               |
| 72     | 4    | uint32_t | Data download slot size                                                                                                                                                 |
| 76     | 4    | uint32_t | Size of the secure area inside the firmware execution area (meaningful only when the firmware is not fully secure).                                                     |
| 80     | 4    | uint32_t | SRAM2 erasing in case of reset                                                                                                                                          |
| 84     | 4    | uint32_t | SRAM2 ECC management activation                                                                                                                                         |
| 88     | 4    | uint32_t | Product state minimal allowed (0x0000ED00, Open / 0x00001700, Provisioning / 0x00002E00, Provisioned / 0x0000C600, TZ-Closed / 0x00007200, Closed / 0x00005C00, Locked) |
| 92     | 70   | array    | Encryption private key (MCUboot format)                                                                                                                                 |
| 162    | 2    | array    | Reserved                                                                                                                                                                |
| 164    | 91   | array    | Authentication public key (MCUboot format)                                                                                                                              |
| 255    | 1    | uint8_t  | Internal tampers activation <sup>(1)</sup>                                                                                                                              |
| 256    | 12   | array    | Reserved (global alignment of OBKeys data on 16 bytes)                                                                                                                  |

1. This is applicable from the SFSP version 2.7.0 on STM32H573xx products, the SFSP version 1.3.0 on STM32H533xx products, the SFSP version 1.3.0 on STM32H553xx products, and SFSP version 1.1.0 on STM32H5F5xx products. For previous SFSP version, when a tamper fault occurs, the system resets in loop until a backup domain power-on reset is performed. Refer to document [7] to get all the information about SFSP version tracking.

## Appendix D STiRoT Data.obk file format

STiRoT\_Data.obk is part of the provisioning process to initialize the images version antirollback counters (zero value) and code image SHA256 reference (zero value).

**Note:** *Be careful to endianness: `uint32_t` 0x01234567 is represented in the obk file as 0x67452301.*

**Table 10. STiRoT\_Data.obk format**

| Offset | Size | Type     | Description                                                                                                          |
|--------|------|----------|----------------------------------------------------------------------------------------------------------------------|
| 0      | 4    | uint32_t | sdp command header : destination address                                                                             |
| 4      | 4    | uint32_t | sdp command header : data size (number of bytes)                                                                     |
| 8      | 4    | uint32_t | sdp command header : encryption enabled/disabled (1/0)                                                               |
| 12     | 32   | array    | SHA256 of the following data content (from offset 44 to 108)                                                         |
| 44     | 4    | uint32_t | Version of the code image, automatically updated during the secure boot and secure firmware update process.          |
| 48     | 4    | uint32_t | Version of the previous code image, automatically updated during the secure boot and secure firmware update process. |
| 52     | 4    | uint32_t | Version of the data image, automatically updated during the secure boot and secure firmware update process.          |
| 56     | 4    | uint32_t | Version of the previous data image, automatically updated during the secure boot and secure firmware update process. |
| 60     | 32   | array    | SHA256 of the code image installed, automatically updated during the secure boot and secure firmware update process. |
| 92     | 16   | array    | Reserved                                                                                                             |

## Appendix E Code init image XML file

**Table 11. Firmware init image generation**

| Parameter                     | Updatable | Description                                                                                                                                                                                                                                                                                                        |
|-------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication key            | Automatic | Private key inherited from STiRoT_Config.xml file.                                                                                                                                                                                                                                                                 |
| Encryption key                | Automatic | Public key inherited from STiRoT_Config.xml file.<br>This parameter can be disabled (adding <code>&lt;enable&gt;</code> property to 0) in order to generate a nonencrypted image                                                                                                                                   |
| Endianness                    | No        | Little endian.                                                                                                                                                                                                                                                                                                     |
| Firmware area size            | Automatic | Firmware slot size information inherited from STiRoT_Config.xml file.                                                                                                                                                                                                                                              |
| Header size                   | No        | 0x400 bytes.                                                                                                                                                                                                                                                                                                       |
| Padding header                | No        | Padding the header with 0xFF to fulfill the 0x400 bytes.                                                                                                                                                                                                                                                           |
| Write option                  | No        | STiRoT implement the overwrite installation method.                                                                                                                                                                                                                                                                |
| Version                       | Yes       | x.y.z firmware image version.                                                                                                                                                                                                                                                                                      |
| Security counter              | Yes       | Security counter value on which the antirollback feature is relying. It can be set to 'auto' to align the image security counter to image version or it can be set manually to a 32-bits value. The antirollback feature is disabled if the security counter value remains unchanged at each version of the image. |
| Align                         | No        | 16-bytes alignment                                                                                                                                                                                                                                                                                                 |
| Firmware download area offset | Automatic | Used to generate .hex binary file format including the destination address. This parameter is inherited from the STiRoT_Config.xml file with the addition of 0x8000000.                                                                                                                                            |
| Firmware binary input file    | Automatic | Location of the firmware binary file. This parameter is updated during the provisioning process based on info from the env.bat file.                                                                                                                                                                               |
| Image output file             | Yes       | Location of the signed image generated. If changed, provisioning scripts must be updated accordingly.                                                                                                                                                                                                              |

## Appendix F Code image XML file

**Table 12. Firmware image generation**

| Parameter                     | Updatable | Description                                                                                                                                                                                                                                                                                                        |
|-------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication key            | Automatic | Private key inherited from STiRoT_Config.xml file.                                                                                                                                                                                                                                                                 |
| Encryption key                | Automatic | Public key inherited from STiRoT_Config.xml file.<br>This parameter can be disabled (adding <code>&lt;enable&gt;</code> property to 0) in order to generate a nonencrypted image                                                                                                                                   |
| Image magic                   | No        | Magic value identifying a firmware code image.                                                                                                                                                                                                                                                                     |
| Endianness                    | No        | Little endian.                                                                                                                                                                                                                                                                                                     |
| Padding                       | No        | Add an installation magic value at the end of the image to trigger image installation.<br>Padding with 0xFF when required.                                                                                                                                                                                         |
| Firmware area size            | Automatic | Firmware slot size information inherited from STiRoT_Config.xml file.                                                                                                                                                                                                                                              |
| Header size                   | No        | 0x400 bytes.                                                                                                                                                                                                                                                                                                       |
| Padding header                | No        | Padding the header with 0xFF to fulfil the 0x400 bytes.                                                                                                                                                                                                                                                            |
| Dependency with data image    | Yes       | Dependency with a specific data image identified with its version. This parameter can be removed by setting the <code>&lt;enable&gt;</code> property to 1.                                                                                                                                                         |
| Write option                  | No        | STiRoT implement the overwrite installation method.                                                                                                                                                                                                                                                                |
| Version                       | Yes       | x.y.z firmware image version.                                                                                                                                                                                                                                                                                      |
| Security counter              | Yes       | Security counter value on which the antirollback feature is relying. It can be set to 'auto' to align the image security counter to image version or it can be set manually to a 32-bits value. The antirollback feature is disabled if the security counter value remains unchanged at each version of the image. |
| Align                         | No        | 16-bytes alignment                                                                                                                                                                                                                                                                                                 |
| Firmware download area offset | Automatic | Used to generate .hex binary file format including the destination address. This parameter is inherited from the STiRoT_Config.xml file with the addition of 0x8000000.                                                                                                                                            |
| Firmware binary input file    | Automatic | Location of the firmware binary file. This parameter is updated during the provisioning process based on info from the env.bat file.                                                                                                                                                                               |
| Image output file             | Yes       | Location of the signed and encrypted image generated. If changed, provisioning scripts must be updated accordingly.                                                                                                                                                                                                |

## Appendix G Data init image XML file

**Table 13. Data init image generation**

| Parameter                 | Updatable | Description                                                                                                                                                                                                                                                                                                        |
|---------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication key        | Automatic | Private key inherited from the STiRoT_Config.xml file.                                                                                                                                                                                                                                                             |
| Encryption key            | Automatic | Public key inherited from the STiRoT_Config.xml file.<br>This parameter can be disabled (adding <code>&lt;enable&gt;</code> property to 0) in order to generate a nonencrypted image                                                                                                                               |
| Endianness                | No        | Little endian.                                                                                                                                                                                                                                                                                                     |
| Data download slot size   | No        | Data download slot size information inherited from the STiRoT_Config.xml file.                                                                                                                                                                                                                                     |
| Header Size               | No        | 0x400 bytes.                                                                                                                                                                                                                                                                                                       |
| Padding Header            | No        | Padding the header with 0xFF to fulfill the 0x400 bytes.                                                                                                                                                                                                                                                           |
| Write option              | No        | STiRoT implement the overwrite installation method.                                                                                                                                                                                                                                                                |
| Version                   | Yes       | x.y.z data image version.                                                                                                                                                                                                                                                                                          |
| Security counter          | Yes       | Security counter value on which the antirollback feature is relying. It can be set to 'auto' to align the image security counter to image version or it can be set manually to a 32-bits value. The antirollback feature is disabled if the security counter value remains unchanged at each version of the image. |
| Align                     | No        | 16-bytes alignment.                                                                                                                                                                                                                                                                                                |
| Data download area offset | Automatic | Used to generate .hex binary file format including the destination address. This parameter is inherited from the STiRoT_Config.xml file with the addition of 0x8000000.                                                                                                                                            |
| Data binary input file    | Automatic | Location of the data binary file. This parameter is updated during the provisioning process based on information from the env.bat file.                                                                                                                                                                            |
| Image output file         | Yes       | Location of the signed image generated. If changed, provisioning scripts must be updated accordingly.                                                                                                                                                                                                              |

## Appendix H Data image XML file

**Table 14. Data image generation**

| Parameter                      | Updatable | Description                                                                                                                                                                                                                                                                                                        |
|--------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication key             | Automatic | Private key inherited from the STiRoT_Config.xml file.                                                                                                                                                                                                                                                             |
| Encryption key                 | Automatic | Public key inherited from the STiRoT_Config.xml file.<br>This parameter can be disabled (adding <code>&lt;enable&gt;</code> property to 0) in order to generate a nonencrypted image                                                                                                                               |
| Image magic                    | No        | Magic value identifying a data image.                                                                                                                                                                                                                                                                              |
| Endianness                     | No        | Little endian.                                                                                                                                                                                                                                                                                                     |
| Padding                        | No        | Add an installation magic value at the end of the image to trigger image installation.<br>Padding with 0xFF when required.                                                                                                                                                                                         |
| Data download slot size        | No        | Data download slot size information inherited from the STiRoT_Config.xml file.                                                                                                                                                                                                                                     |
| Header Size                    | No        | 0x400 bytes.                                                                                                                                                                                                                                                                                                       |
| Padding Header                 | No        | Padding the header with 0xFF to fulfill the 0x400 bytes.                                                                                                                                                                                                                                                           |
| Dependency with firmware image | Yes       | Dependency with a specific firmware image identified with its version. This parameter can be removed by setting the <code>&lt;enable&gt;</code> property to 1.                                                                                                                                                     |
| Write option                   | No        | STiRoT implement the overwrite installation method.                                                                                                                                                                                                                                                                |
| Version                        | Yes       | x.y.z data image version.                                                                                                                                                                                                                                                                                          |
| Security counter               | Yes       | Security counter value on which the antirollback feature is relying. It can be set to 'auto' to align the image security counter to image version or it can be set manually to a 32-bits value. The antirollback feature is disabled if the security counter value remains unchanged at each version of the image. |
| Align                          | No        | 16-bytes alignment.                                                                                                                                                                                                                                                                                                |
| Data download area offset      | Automatic | Used to generate .hex binary file format including the destination address. This parameter is inherited from the STiRoT_Config.xml file with the addition of 0x8000000.                                                                                                                                            |
| Data binary input file         | Automatic | Location of the data binary file. This parameter is updated during the provisioning process based on information from the env.bat file.                                                                                                                                                                            |
| Image output file              | Yes       | Location of the signed and encrypted image generated. If changed, provisioning scripts must be updated accordingly.                                                                                                                                                                                                |

## Appendix I STiRoT execution status

STiRoT is executed as a black box, but there is a way to verify the steps that have been executed and in case of an issue where the STiRoT execution has stopped.

For each HDPL OBKeys, there are four bytes reserved for the status log. This information can be extracted using the debug Authentication discovery command (refer to [5] for more information).

**Table 15. Execution status**

| Executed step                 | Status coding | Description                                                                                                     |
|-------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------|
| STIROT_INIT_DONE              | 0x00000001UL  | Device initialization done. STiRoT bootpath selected. STiRoT executed.                                          |
| STIROT_CONFIG_DONE            | 0x00000002UL  | STiRoT configuration according to STiRoT_Config.obk is verified to be correct in OBKeys HDPL1.                  |
| STIROT_SECURITY_DONE          | 0x00000004UL  | Security activated. OBKeys STiRoT configuration verified to be correct.                                         |
| STIROT_SLOT_PRIMARY_1_VALID   | 0x00000010UL  | Authenticity and integrity verified for the user application firmware image in the execution slot.              |
| STIROT_SLOT_PRIMARY_2_VALID   | 0x00000020UL  | Authenticity and integrity verified for the data image in the HDPL2 OBKeys secure storage.                      |
| STIROT_SLOT_SECONDARY_1_VALID | 0x00000040UL  | Authenticity and integrity verified for the user application firmware image in the download slot.               |
| STIROT_SLOT_SECONDARY_2_VALID | 0x00000080UL  | Authenticity and integrity verified for the data image in the download slot.                                    |
| STIROT_PARSING_DONE           | 0x00000100UL  | Slots parsing done.                                                                                             |
| STIROT_INSTALLATION_DONE      | 0x00000200UL  | Image(s) in the download slot (one or two images depending on config) are installed.                            |
| STIROT_VALIDATION_DONE        | 0x00000400UL  | Execution slots analysis completed to verify which image is valid (integrity and authenticity).                 |
| STIROT_JUMP_BL                | 0x00001000UL  | At least one image is not valid -> bootloader execution.                                                        |
| STIROT_JUMP_APPLI             | 0x00002000UL  | All images (one or two depending on config) are valid -> application execution.                                 |
| STIROT_NOJUMP                 | 0x00004000UL  | At least one image is not valid, but bootloader execution is not allowed (option defined in STiRoT_Config.xml). |

## Appendix J Troubleshooting

Table 16. Troubleshooting

| Problem                                                | Possible solution                                                            |
|--------------------------------------------------------|------------------------------------------------------------------------------|
| Provisioning errors in ob_flash_programming.bat script | STM32CubeProgrammer is still connected to the device.                        |
| Provisioning errors                                    | Device is not in OPEN product state: regression.bat script must be executed. |



## Revision history

**Table 17. Document revision history**

| Date        | Revision | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21-Sep-2023 | 1        | Initial release.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 19-Mar-2024 | 2        | Terminology updated.<br>Appendix added: Appendix G: Troubleshooting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 20-Mar-2024 | 3        | Topic updated: Section 1: General information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 08-Jan-2025 | 4        | Section 2.4: STiRoT constraints added.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 03-Mar-2025 | 5        | Appendix D: STiRoT Data.obk file format added.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 13-Jun-2025 | 6        | Updated: <ul style="list-style-type: none"> <li>Section 2.4: STiRoT constraints</li> <li>Figure 6. STiRoT provisioning process</li> <li>Section 4: OBKeys configuration file</li> <li>Section 5: Images generation</li> <li>Figure 7. Firmware image generation</li> <li>Figure 8. Data image generation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                     |
| 17-Dec-2025 | 7        | Added: <ul style="list-style-type: none"> <li>Section 5.1: Initial firmware image generation</li> <li>Section 5.3: Initial data image generation</li> <li>Appendix E: Code init image XML file</li> <li>Appendix G: Data init image XML file</li> </ul> Updated: <ul style="list-style-type: none"> <li>Section 2.1: Overview</li> <li>Section 2.4: STiRoT constraints</li> <li>Section 3: Provisioning process</li> <li>Section 5.2: Update firmware image generation</li> <li>Section 5.4: Update data image generation</li> <li>Appendix F: Code image XML file</li> <li>Appendix H: Data image XML file</li> <li>Section 4: OBKeys configuration file</li> <li>Appendix C: STiRoT Config.obk file format</li> </ul> |
| 11-Feb-2026 | 8        | Added STM32H5F5xx part number to the document                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 15-Jun-2026 | 9        | Added STM32H553xx part number to the document                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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